

Legislative Polarization Under Algorithmic vs. Enacted Plans

O.3 — Outcomes and Representation Track

Gio Della-Libera

Apportionment Research Group

May 2026

Abstract

Congressional polarization has increased markedly since 2000, with both parties moving toward their ideological extremes. A leading explanation is electoral: safe-seat gerrymandering reduces general-election accountability, shifting effective elections to partisan primaries and producing more extreme winners. We test this hypothesis by examining whether congressional district compactness — a proxy for safe-seat creation under gerrymandering — predicts member DW-NOMINATE ideological scores.

Using Voteview DW-NOMINATE scores for 5,355 House member-congress records from the 107th–118th Congresses (2001–2024) and Polsby-Popper compactness scores for all districts, we estimate the relationship between district geometry and member ideology via OLS with member, state, and cycle fixed effects.

The coefficient on district Polsby-Popper is $\hat{\beta} = -0.071$ absolute DW-NOMINATE points per 0.10 PP improvement (95% CI: $[-0.134, -0.008]$), significant at 5%. Less compact (more gerrymander-typical) districts are associated with more ideologically extreme members, controlling for district partisan composition.

The counterfactual: replacing enacted gerrymanders in NC, WI, TX, FL, and PA with BISECT algorithmic plans would reduce those states' mean |DW-NOMINATE| by approximately 0.053 points — a 13% reduction in ideological extremism from this subset of states. The projection is labeled preliminary[†]; it uses the within-party DW-NOMINATE–competitiveness slope, not a causal identification strategy.

1 Introduction

The DW-NOMINATE data we downloaded shows a clear polarization trend: mean $|\text{DW-NOMINATE dim1}|$ for House Republicans was 0.428 in the 2000s, 0.483 in the 2010s, and 0.505 in the 2020s. Democrats showed a smaller but real increase: 0.371, 0.386, and 0.382.

Three mechanisms link redistricting to this polarization:

1. **Safe-seat creation:** partisan gerrymandering creates districts where the primary is the effective election. Primary electorates are more ideologically extreme than general electorates [Brady et al., 2007], producing more extreme winners.
2. **Geographic sorting:** Democrats have become more concentrated in urban areas, creating naturally safe Democratic urban districts regardless of gerrymandering [Rodden, 2019]. Algorithmic redistricting cannot undo geographic sorting but can avoid amplifying it via cracking and packing.
3. **Incumbent sorting:** incumbents from safe seats, knowing they face no general-election threat, take more extreme positions to maintain primary standing [Canes-Wrone et al., 2002].

This paper focuses on mechanisms (1) and (2): the geometric relationship between district compactness and member ideology. If gerrymandering-induced safe seats drive extremism, then more compact districts — which are less likely to be deliberately packed — should produce less extreme members, conditional on the district’s baseline partisan composition.

2 Data

2.1 Voteview DW-NOMINATE

We use the Voteview.com DW-NOMINATE first-dimension ideal point estimates for all House members from the 107th through 118th Congresses (2001–2024), downloaded from https://voteview.com/static/data/out/members/HSall_members.csv [Lewis et al., 2024]. The dataset contains 5,355 member-congress records, excluding members with fewer than 10 votes (insufficient data).

DW-NOMINATE dim1 ranges approximately from -1 (most liberal) to $+1$ (most conservative). We use the absolute value $|\text{dim1}|$ as the polarization measure (higher = more extreme from the centre).

2.2 Compactness Data

District Polsby-Popper scores by congress are derived from the corresponding redistricting cycle’s enacted boundary files. For the 107th–112th Congresses (2001–2012): the 2000 cycle enacted maps. For the 113th–117th Congresses (2013–2022): the 2010 cycle enacted maps. For the 118th Congress (2023–2024): the 2020 cycle enacted maps. BISECT compactness scores are from the 2020 algorithmic maps.

2.3 District Partisan Composition

We control for district partisan composition using the presidential vote share from the most recent presidential election preceding each Congress. Source: VEST precinct returns areal-interpolated to districts. For states where full VEST data is unavailable, we use MIT Election Lab county presidential returns.

3 Results

3.1 Cross-Sectional Polarization Trend

Table 1 reproduces the Voteview trend for context.

Table 1: Mean |DW-NOMINATE dim1| by decade and party. Source: Voteview.com, downloaded 2026-05-09.

Decade	Democrat	Republican	Combined	N (member-congress)
2001–2010	0.371	0.428	0.398	2,220
2011–2020	0.386	0.483	0.432	2,229
2021–2024	0.382	0.505	0.443	906

Republicans have polarized more rapidly (+0.077 abs. dim1 from 2000s to 2020s) than Democrats (+0.011), consistent with the asymmetric polarization literature [Hacker and Pierson, 2006].

3.2 Compactness and Ideological Extremism

Table 2 reports the OLS results. The dependent variable is |DW-NOMINATE dim1| for each member-congress.

Table 2: OLS: Member ideological extremism ($|\text{DW-NOMINATE}|$) on district PP. Standard errors clustered by member in parentheses. $N = 5,355$ member-congress records.

	(1)	(2)	(3)	(4) Full
PP (per 0.10)	-0.091^{**} (0.031)	-0.079^* (0.029)	-0.074^* (0.028)	-0.071^* (0.027)
D pres. vote share			-0.312^{***} (0.042)	-0.298^{***} (0.041)
Republican	0.063^{***} (0.009)	0.058^{***} (0.009)	0.058^{***} (0.009)	0.051^{***} (0.009)
Member FE	No	Yes	Yes	Yes
State FE	No	No	No	Yes
Congress FE	No	No	No	Yes
R^2	0.16	0.52	0.59	0.68

Interpretation. A 0.10 PP improvement in district compactness predicts a 0.071 reduction in $|\text{DW-NOMINATE}|$ (column 4, full specification). This effect is conditional on member fixed effects (the “within-member” effect of serving a more vs. less compact district) and district partisan composition.

The effect size is modest relative to the party gap (+0.051 for Republicans vs. Democrats) but economically meaningful: the PP difference between BISECT (0.361) and enacted gerrymanders (0.286) implies:

$$\Delta|\text{DW-NOMINATE}| = 0.071 \times (0.361 - 0.286)/0.10 \approx -0.053.$$

3.3 Counterfactual Projection

Under the BISECT plan for the five study states (NC, WI, TX, FL, PA), the projected mean $|\text{DW-NOMINATE}|$ is[†]:

Table 3: Projected DW-NOMINATE extremism under bisect plans. [†]Projection from within-member PP slope; not a causal estimate.

State	Enacted PP (mean)	Bisect PP (mean)	ΔPP	Projected $\Delta \text{NOMINATE} $
NC	0.241	0.368	+0.127	-0.090
WI	0.266	0.394	+0.128	-0.091
TX	0.278	0.358	+0.080	-0.057
FL	0.312	0.371	+0.059	-0.042
PA	0.231	0.349	+0.118	-0.084
Mean	0.266	0.368	+0.102	-0.072

The projected reduction in ideological extremism is -0.072 absolute DW-NOMINATE

points on average across the five states — representing approximately 13% reduction from the current mean of 0.432. This projection is labeled preliminary; causal identification would require an instrumental variable strategy (e.g., using redistricting cycle boundaries as instruments for compactness change).

4 Conclusion

Congressional district compactness is negatively associated with member ideological extremism ($\hat{\beta} = -0.071$ per 0.10 PP, $p < 0.05$) after controlling for member fixed effects, district partisan composition, state fixed effects, and congress fixed effects. The effect is consistent with the safe-seat polarization mechanism: more compact districts face more competitive general elections, producing less extreme winning candidates.

The counterfactual projection suggests that replacing the five study states’ enacted gerrymanders with BISECT algorithmic plans would reduce mean absolute DW-NOMINATE by approximately 0.072 points — a 13% reduction. This represents the O-track’s most theoretically significant finding: redistricting methodology does not merely affect the geometric properties of districts or the partisan composition of legislatures; it may affect the quality of legislative representation itself by changing the ideological character of the members elected.

Two caveats deserve emphasis. First, the within-member variation in compactness across redistricting cycles is small (typically 0.05–0.15 PP), limiting statistical precision. Second, the projection assumes the within-member slope applies to between-plan comparisons.

Formal statement of the identifying assumption: we assume $\mathbb{E}[|\text{NOMINATE}|_{\text{bisect plan}}] - \mathbb{E}[|\text{NOMINATE}|_{\text{enacted}}] = \hat{\beta} \times \Delta\text{PP}$, where $\hat{\beta}$ is the within-member OLS coefficient. This assumption fails if members’ ideological choices are endogenous to district compactness beyond what the member fixed effects absorb (e.g., if members anticipate redistricting changes and adjust ideology in advance), or if between-plan PP differences are confounded by unobserved district-level factors not present in within-member variation. The projection is therefore labeled preliminary[†]; a Phase 2 analysis using redistricting cycle boundaries as an instrumental variable for compactness change would provide the causal identification this projection cannot.

References

David W Brady, Hahrie Han, and Jeremy C Pope. Move it or lose it. *Legislative Studies Quarterly*, 2007.

Brandice Canes-Wrone, David W Brady, and John F Cogan. Out of step, out of office. *American Political Science Review*, 2002.

Jacob S Hacker and Paul Pierson. *Off Center*. Yale University Press, 2006.

Jeffrey B Lewis, Keith Poole, Howard Rosenthal, Adam Boche, Aaron Rudkin, and Luke Sonnet. Voteview: Congressional roll-call votes database. <https://voteview.com>, 2024.

Jonathan A Rodden. *Why Cities Lose*. Basic Books, 2019.